Contact Manager Software

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Objectives

Accessibility

Ensure the application is easy to use for first time users, and across different types of platforms

User-Friendly

Ensure a clean navigation for the user to use such as add, edit, view their contact information.

Functional

Include basic features, adding/removing/editing contacts, while maintaining functionality

Cost Estimation

Effort

- Function Points: **102.9 FP**
- Code Size: **~5.7 KLOC**
- Effort: **14.8 Person-Months**

Cost Breakdown

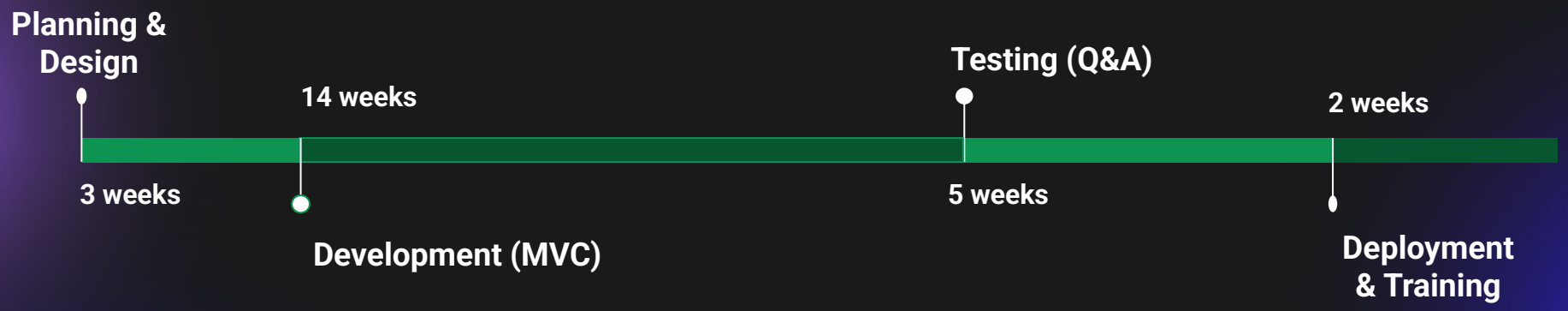
- Personnel: **$142,000**
- Hardware: **$6,500**
- Software: **$2,400**

Final Pricing

- Development Cost: **$125,800**
  - Overhead (20%)
  - Profit (15%)

Total Price: $173,604

Project Timeline



Functional Requirements

- Add, edit, delete, and view contacts
- Blacklist contacts (block calls & SMS)
- Call or send SMS directly from contact
- Create and manage contact groups
- Sort contacts (name / group)
- Import & export contact data

Non-Functional Requirements

Performance

- Load contacts within **2 seconds**
- Call/SMS actions within **3 seconds**

Usability

- Simple, intuitive interface
- Learnable within **15 minutes**

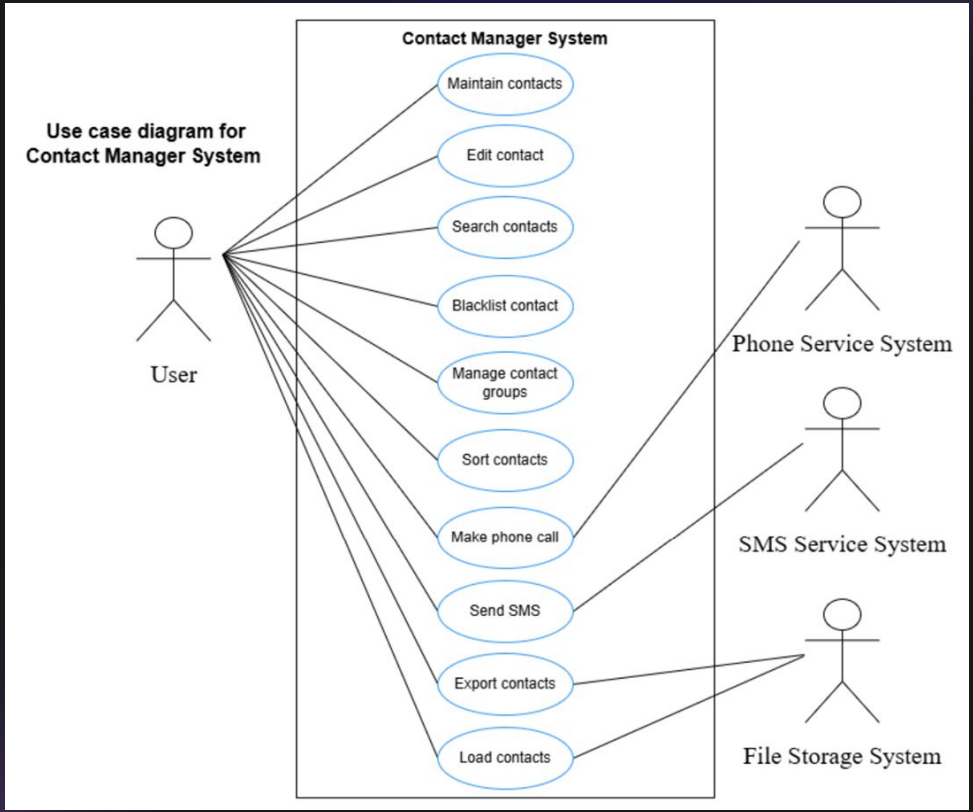
Security & Reliability

- Secure data storage
- No data loss
- Input validation

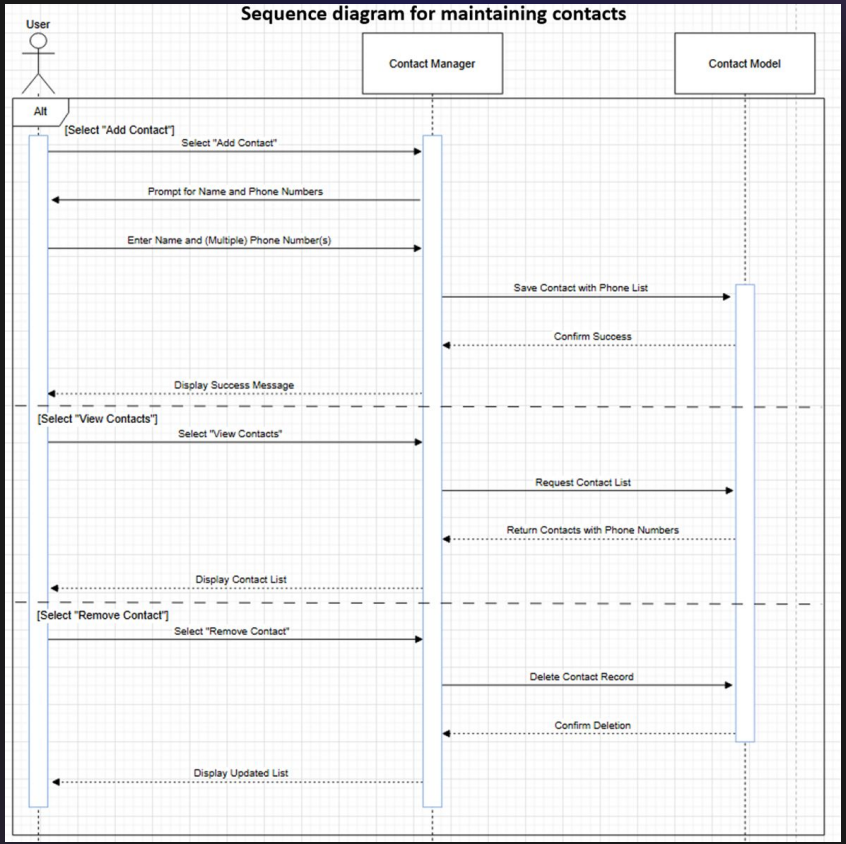
Compatibility

- Works on **Android & iOS**
- Runs on standard devices

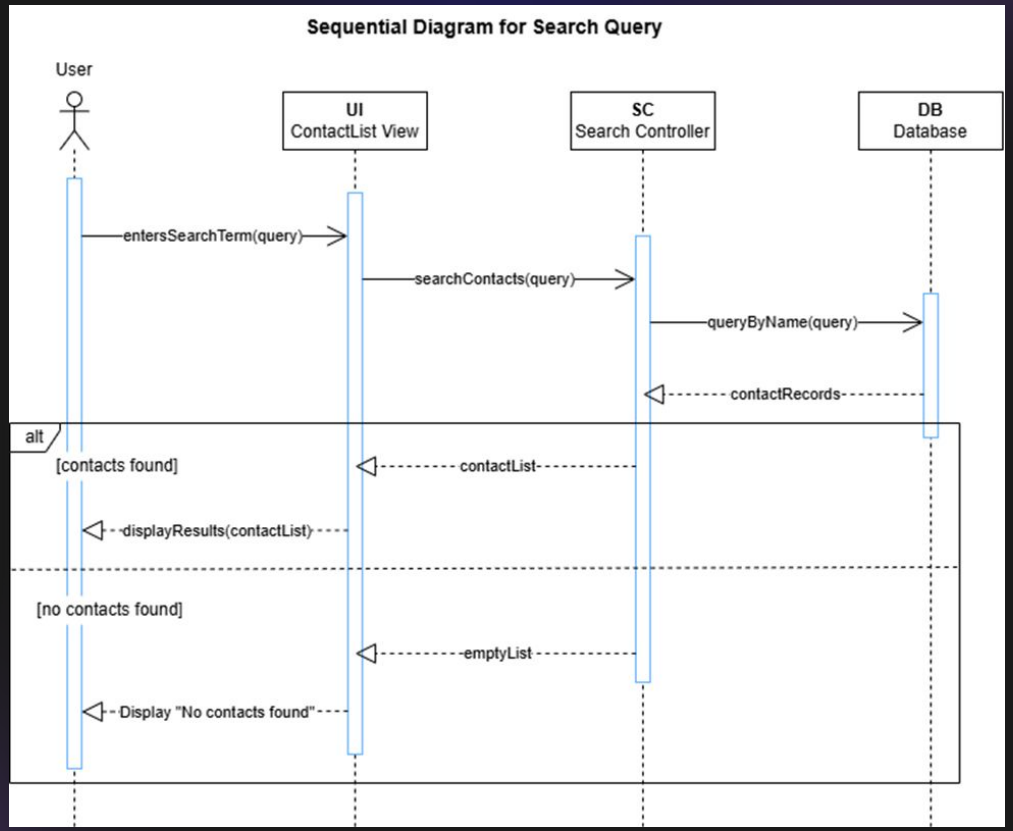
Use Case Diagram



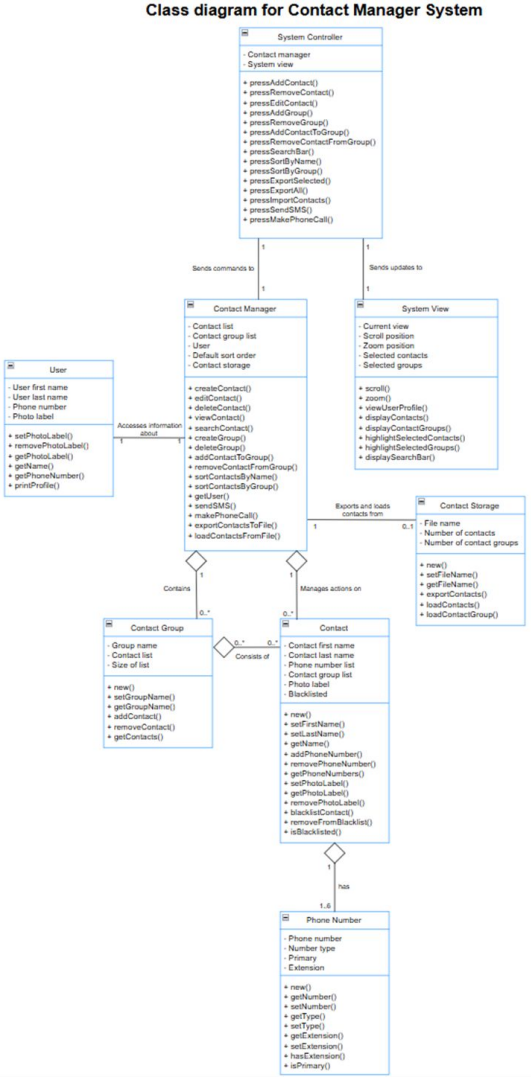
Sequence Diagram for Maintaining Contacts



Sequence Diagram for Search Query



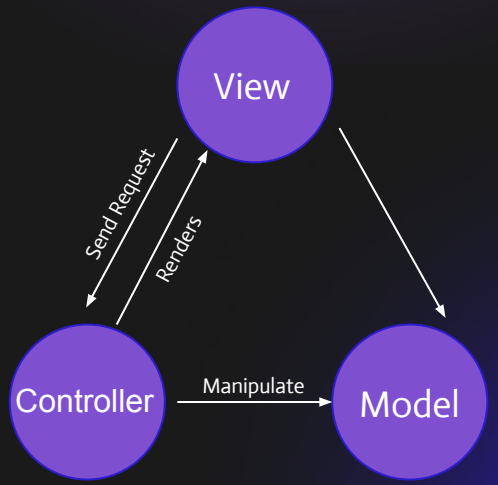
Class Diagram



Architectural design

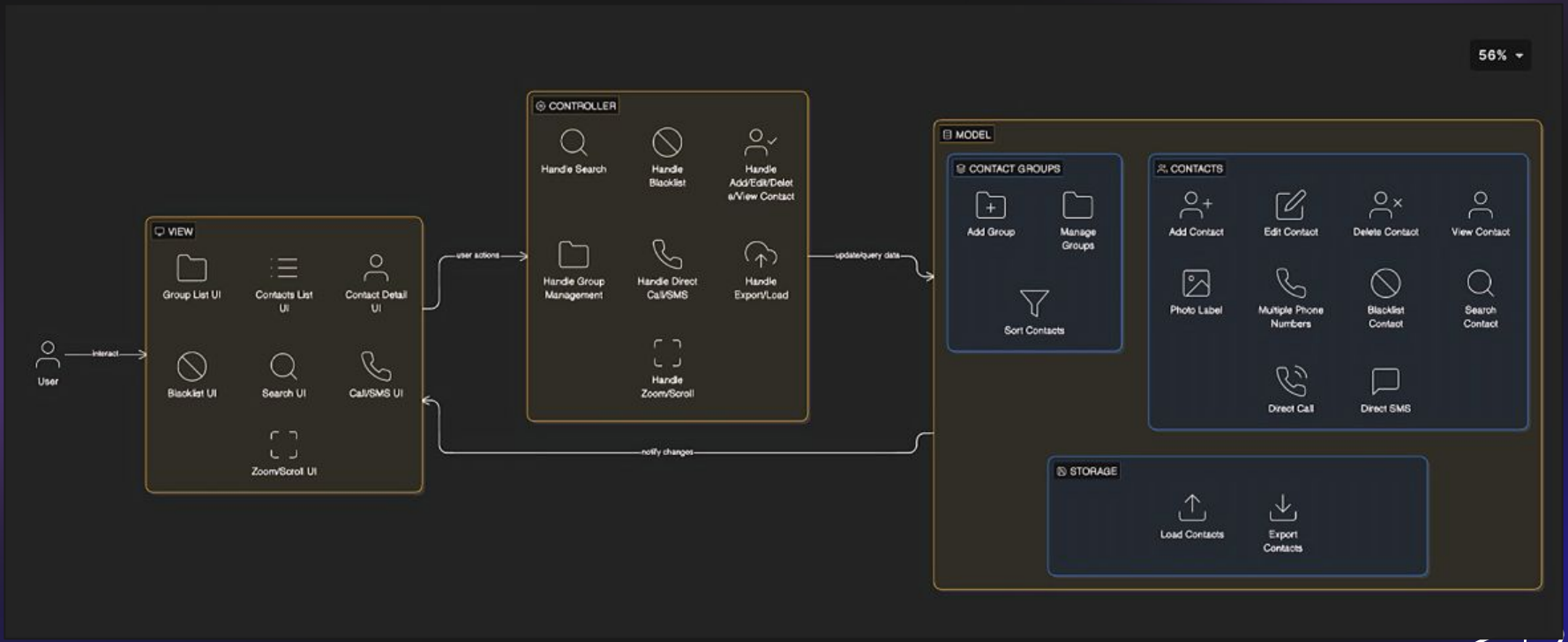
Model-View-Controller (MVC)

- Divides the system structure into 3 logical components which function together
  1. **Model** (Core Data and Logic)
     - Manages contact data, sorting algorithms, and grouping.
     - Handles import/export functionality
  2. **View** (User Interface)
     - Renders the data in multiple formats
  3. **Controller** (Interaction Logic)
     - Processes user inputs and coordinates updates between Model and View



- Why MVC?
  - **Independent Representation** : Data change synchronizes across all views
  - **Scalability** : Facilitates future expansion, such as moving to a web-based app
  - **Maintainability** : Ensures fundamental features remain modular and maintainable

Architectural design



Comparison with similar designs

Apps that were studied:

- Apple Contacts
- Android Contacts
- Google Contacts

Similarities:

- Add/View/Edit/delete contacts
- Support multiple phone numbers
- Support adding a photo label for a contact
- Search contacts by contact name
- Blacklist(Block SMS and Phone)
- Directly make phone calls and send SMS from a contact view

Comparison with similar designs

Similarities:

- Add contact Groups
- Manage Contact Groups (Add/Remove contacts)
- Support adding a photo label for a contact
- Zoom in/out
- Scroll whenever necessary

Differences:

- Saving User data :
  - Cloud VS locally

Demonstrate a true and false test-case

```java
public static boolean isValidPhoneNumber(String number) {
    if (number == null || number.trim().isEmpty()) return false;
    String cleaned = number.replaceAll( regex: "[\\s\\-().]+",  replacement: "");
    return cleaned.matches( regex: "^\\+?[0-9]{7,15}$");
}
```

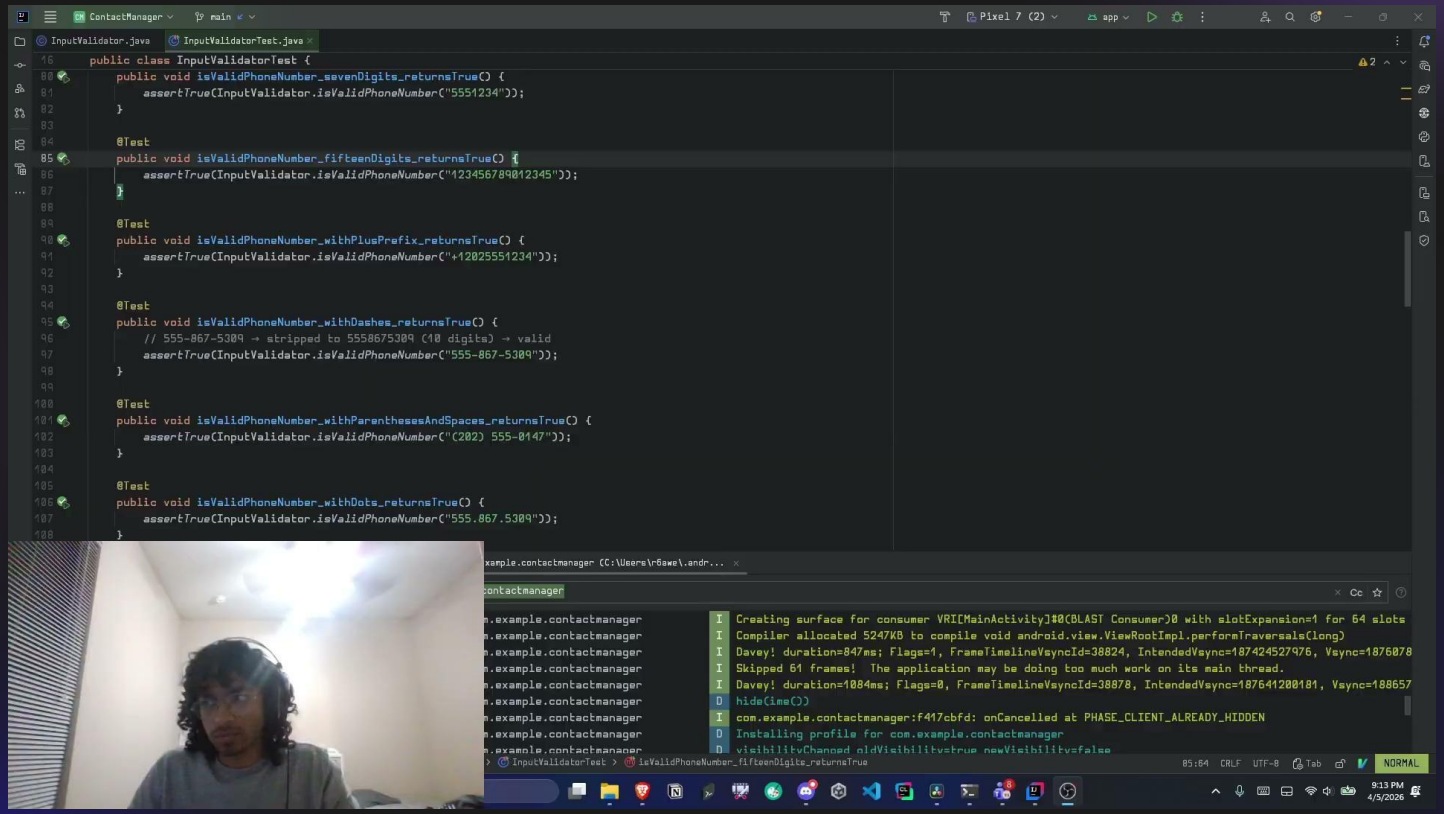
```java
@Test
public void isValidPhoneNumber_withDots_returnsTrue() {
    assertTrue(InputValidator.isValidPhoneNumber("555.867.5309"));
}
```

✓ isValidPhoneNumber_withDots_returnsTrue

```java
@Test
public void isValidPhoneNumber_sixDigits_returnsFalse() {
    assertFalse(InputValidator.isValidPhoneNumber("123456"));
}
```

✓ isValidPhoneNumber_sixDigits_returnsFalse

Prototype of proposed application



Project Reflection

Challenges

- Coordination across multiple team members
- Ensuring consistency across diagrams and report sections
- Expanding from design to realistic planning

Lessons Learned

- Importance of clear task delegation and communication
- Understanding how software engineering concepts apply in practice
- Observing how in depth planning in real-world projects goes

Conclusion

- Successfully developed a complete and realistic software project design
- Strengthened understanding of the software development life cycle

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Thanks!

Q&A